Algorithmic Political Warfare and Democratic Resilience: A Digital Twin of the 2026 Brazilian Presidential Election

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Abstract

Digital communication technologies and closed encrypted messaging networks have transformed contemporary electoral contests into high-velocity asymmetric information environments. In polarized democracies, computational propaganda and targeted outrage blitzes frequently bypass institutional gatekeeping, shifting voting outcomes during narrow pre-election windows. Here we investigate the systemic drivers of late-campaign digital shocks and evaluate active counter-tactics through a calibrated agent-based digital twin of the 2026 Brazilian presidential election. Integrating an urban multiplex interpersonal network, a five-modality traditional mass media ecosystem, and four algorithmic social platforms, our model reproduces longitudinal survey waves and official election tallies with high statistical fidelity ($R^2 = 0.9955$, root-mean-square error 0.85 percentage points). We identify a fundamental cognitive negativity asymmetry whereby aversion acquisition outpaces its decay by $3.5\times$, rendering electorates vulnerable to acute 48-hour disinformation blitzes that overcome baseline political majorities. We demonstrate that conventional fact-checking and legacy broadcast airtime fail to counter these surges. In contrast, operationalizing the counter-tactical doctrine of Janonismo Cultural, namely algorithmic asphyxiation, strict monothematism, material loss-aversion framing, WhatsApp synchronization, mass capital rallies, and continuous live saturation, systematically reverses candidate rejection, crossing the majority threshold to achieve runoff victory. These findings establish that democratic resilience requires the structural operationalization of algorithmic dynamics in defense of popular majorities.

Keywords: Computational social science; Agent-based digital twin; Algorithmic polarization; Political disinformation; Electoral dynamics; Democratic resilience.

Modern democratic elections are increasingly decided within complex digital communication ecosystems that couple face-to-face deliberation, broadcast mass media, and algorithmic social networking platforms [1–3]. Over the past decade, the rapid penetration of mobile smartphones and end-to-end encrypted messaging applications has restructured the flow of political information, creating fertile ground for computational propaganda, astroturfing, and viral moral panics [4–6]. In many democracies, right-wing populist movements have exploited these architectural affordances with remarkable efficacy, deploying asymmetric disinformation campaigns that outpace institutional verification and alter voter preferences during critical late-stage campaign sprints [7–9].

Faced with these computational offensives, traditional political campaigns and institutional democratic actors have predominantly relied on conventional remedies, including fact-checking press releases, judicial moderation petitions, and legacy broadcast advertising [10–12]. Empirical evidence across recent global elections suggests that these defensive postures are structurally inadequate [5, 13]. Fact-checking rebuttals inadvertently trigger cognitive backfire effects among polarized partisans while suffering from an acute temporal disadvantage relative to viral digital cascades [14, 15]. Meanwhile, statutory broadcast airtime on television and radio, historically the central arena of political competition, struggles to penetrate younger and economically precarious demographics whose political worldviews are increasingly forged on algorithmic discovery feeds and dark-social messaging groups [16, 17].

The 2026 Brazilian presidential contest provides an ideal quasi-natural experiment to investigate both the mechanics of late-campaign digital disruption and the conditions under which it can be neutralized. Pitting

incumbent president Luiz Inácio Lula da Silva against Senator Flávio Bolsonaro, the contest unfolded across two sharply contrasting strategic regimes. Throughout the first round, the incumbent coalition maintained a conventional, institutional communication posture backed by a $+27.3\%$ advantage in statutory television airtime. Despite steady polling leads throughout the 50-day campaign, the incumbent was overtaken on election eve by an acute, coordinated multi-channel blitz that lifted Flávio Bolsonaro into first place with 47.03% of the valid vote.

In the ensuing second-round runoff, the progressive campaign executed a radical strategic pivot by adopting the counter-tactical doctrine of *Janonismo Cultural*, codified by Federal Deputy André Janones [18]. Rejecting passive fact-checking, this operational doctrine deployed aggressive algorithmic counter-offensives designed to invert platform curation mechanics, choke opponent virality, and mobilize working-class turnout. This counter-offensive culminated in an unprecedented electoral turnaround, lifting Lula to victory with 51.79% of the vote and marking the first time in modern Brazilian democratic history that an executive contender trailing in the first round won the presidency in the runoff.

Here, we construct a high-resolution computational digital twin of the 2026 Brazilian electorate to systematically decipher these dynamics. Grounded in computational epidemiology and multi-layer network physics [19–21], our framework integrates three interacting influence substrates, comprising an urban multiplex interpersonal network calibrated to national census demography [22], a five-modality traditional mass media ecosystem [16], and a multi-platform algorithmic social media network [17]. By parameterizing cognitive trait heterogeneity, dual-timescale conviction crystallization, and asymmetric candidate rejection dynamics, we track the macro-electoral consequences of microscopic information flows.

Through extensive Monte Carlo ensembles ($N = 100{,}000$ citizens across 100 independent realizations), we resolve two fundamental questions. First, what structural vulnerabilities in human psychology and digital networks enable late-campaign disinformation blitzes to overturn established political majorities? Second, can progressive democratic actors defeat computational propaganda without resorting to state censorship, and what specific algorithmic mechanisms determine the success of digital counter-offensives?

Results

The 2026 Brazilian presidential election in empirical data

Brazil operates as a presidential federative republic with concurrent general elections held every four years, allowing executive incumbents a single consecutive reelection alongside the renewal of the bicameral National Congress. In the 2026 contest, incumbent president Luiz Inácio Lula da Silva of the Workers' Party (*Partido dos Trabalhadores*, PT) sought reelection, challenged primarily by Senator Flávio Bolsonaro of the right-wing Liberal Party (*Partido Liberal*, PL). Flávio is the eldest son of former president Jair Bolsonaro, who had narrowly lost the 2022 election to Lula and was subsequently incarcerated following judicial convictions related to an attempted coup d'état. Third-party contenders exerted minimal electoral resonance, consisting largely of conservative figures attempting to capture the Bolsonarista electorate while Flávio's ability to unify the right remained unproven.

The 2026 presidential race was defined by two sharply divergent campaign regimes across its first and second rounds (Fig. 1). Throughout the opening 50-day campaign, voter intentions exhibited remarkable inertia (Fig. 1b): longitudinal tracking surveys by Datafolha depicted a steady, narrow lead for Lula (fluctuating between 41.8% and 45.0% of valid votes) over Flávio (37.5% to 41.0%), with third-way candidates holding 14.0% to 22.0%. During this period, candidate rejection was mutually crystallized in a narrow band of 45%–47% (Fig. 1c). Neither front-runner participated in televised debates, and public discourse was largely overshadowed by controversies concerning prominent judiciary figures rather than substantive executive policy.

This opening phase also reflected sharp structural asymmetries in institutional campaign endowments. Under Brazilian electoral law (Lei nº 9.504/1997), broadcast airtime on free television and radio (HGPE) is distributed proportionally to coalition representation in the Chamber of Deputies (Fig. 1d): Lula's broad seven-party coalition secured 5m 31s per daily block (44.1% of total airtime; 433 30-second insertions) compared to Flávio's 4m 20s (34.7%; 339 insertions), yielding a $+27.3\%$ airtime advantage for the incumbent. Financially, while the PL controlled the largest national allocation from the public Special Campaign Financing Fund

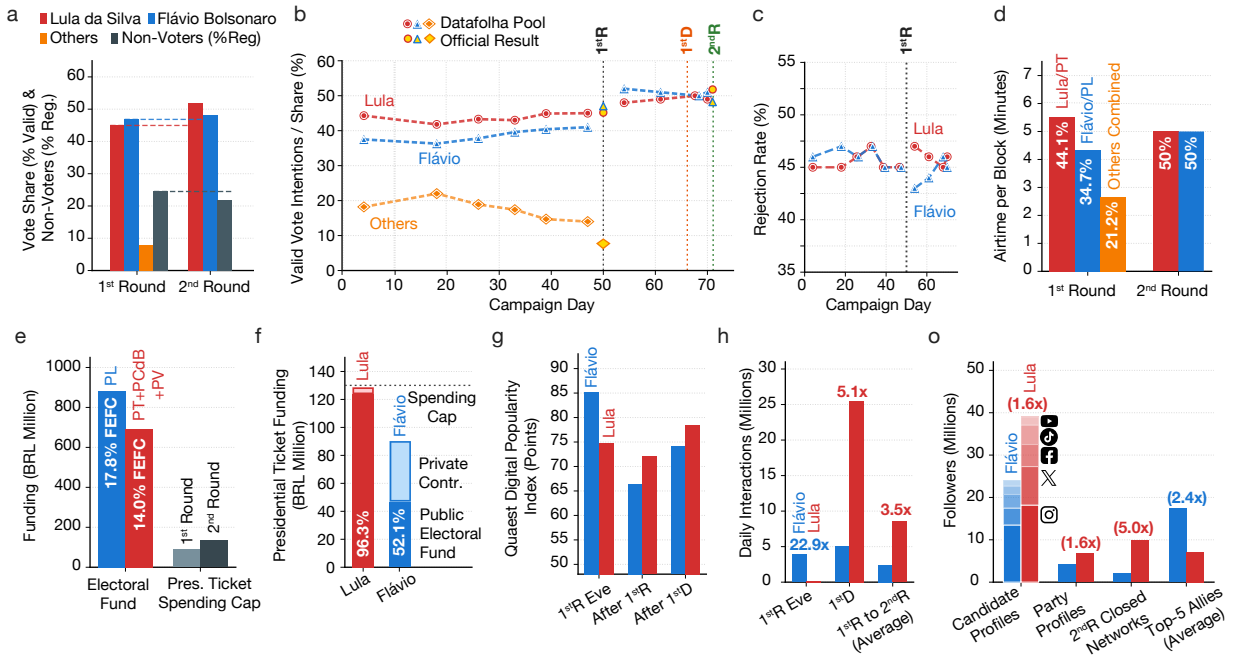


Figure 1: **Fig. 1 | The 2026 Brazilian presidential election in empirical data. a**, Official Superior Electoral Court (TSE) election results for Round 1 (October 4) and Round 2 (October 25), alongside election-day turnout mobilization that compressed non-voters by -2.91 percentage points. **b**, Longitudinal tracking of valid vote intentions across six Round 1 and four Round 2 Datafolha tracking surveys, with gold markers indicating official ballot tallies. **c**, Candidate rejection dynamics (Datafolha: percentage stating they would not vote for the candidate under any circumstance), highlighting the second-round reversal that drove Flávio's rejection to 46.0%. **d**, Statutory free broadcast airtime (HGPE) per daily block on television and radio, shifting from a $+27.3\%$ coalition advantage for Lula in Round 1 to strict legal parity (5m 00s each) in Round 2. **e**, Macro public party financing (FEFC; total national pool R$ 4.96 billion) and statutory TSE presidential spending ceilings (R$ 88.9M in Round 1, R$ 133.4M cumulative in Round 2). **f**, Presidential ticket funding profiles, comparing Lula's state-financed budget (96.3% FEFC) with Flávio's hybrid apparatus that secured a $9.0\times$ private donation advantage. **g**, Quaest Digital Popularity Index (IPD; 0–100 scale, 175 metrics) across three milestone waves, illustrating a $+16.0$-point net turnaround swing for Lula. **h**, Daily social interaction volumes (millions of interactions per day, Bites data) during the Round 1 shock, televised debate, and the 13-day runoff window. **i**, Follower ecosystem and bench architecture, reporting candidate direct accounts across five platforms, official party handles, closed WhatsApp groups (Exército do Lula 10.0M vs. Tropa do Flávio 2.0M), and the arithmetic mean of the top-5 legislative surrogates per coalition.

(FEFC; total national pool R$ 4.96 billion; Supplementary Note 4), presidential spending operated under statutory ceilings (R$ 88.9 million in Round 1, expanding to R$ 133.4 million in Round 2; Fig. 1e). The two tickets diverged sharply in revenue architecture (Fig. 1f): Lula deployed a state-financed apparatus funded 96.3% by public campaign funds (R$ 123.8M of R$ 128.6M declared), whereas Flávio relied on a hybrid structure raising an unprecedented R$ 43.2M in private contributions and crowdfunding, achieving a $9.0\times$ private capital advantage over Lula (R$ 4.8M).

The first-round ballot on October 4 produced a profound empirical shock (Fig. 1a). Flávio Bolsonaro led with 47.03% of the valid vote (55.7 million votes), finishing ahead of Lula (45.16%, 53.5 million votes) as third-party support collapsed to 7.81% (9.3 million votes). Concurrently, voter non-participation reached a historical high of 24.70% (20.90% abstention and 4.80% blank/null ballots; Fig. 1a). On election night, Bolsonarista digital networks mobilized aggressively, generating 3.90 million daily interactions compared to Lula's 0.17 million, representing a $22.9\times$ engagement advantage (Fig. 1h).

This outcome prompted an immediate realignment. Initial post-election surveys projected a decisive advantage for Flávio (52.0% vs. 48.0% on October 8; Fig. 1b), supported by a post-victory honeymoon that

lowered his rejection rate to 43.0% while Lula's rose to 47.0% (Fig. 1c), mirrored by a 9.3-point lead for Flávio in the Quaest Digital Popularity Index (80.0 vs. 70.7; Fig. 1g). Under runoff rules, broadcast airtime was equalized to strict 5m 00s parity (25 spots/day each; Fig. 1d), shifting the decisive battleground to the digital sphere.

In response, the second-round campaign adopted a restructured tactical posture. Although Flávio possessed a $2.44\times$ multiplier in legislative surrogate bench strength (averaging 17.4 million cross-platform followers across his top-5 allies, led by Nikolas Ferreira with 52.0M, versus 7.1 million for PT allies; Fig. 1i), Lula counterbalanced this with a larger personal base (39.0M vs. 24.1M) and a massive closed WhatsApp infrastructure (Exército do Lula, 10.0 million participants vs. Flávio's 2.0 million; Fig. 1i). Orchestrated around aggressive responsive messaging and material-loss framing regarding welfare protections, the Lulista digital counter-offensive dominated engagement during the October 16 televised debate (25.5M vs. 5.0M interactions; Fig. 1h) and sustained a $3.5\times$ daily interaction advantage throughout the runoff (8.52M vs. 2.41M/day; Fig. 1h).

This digital saturation steadily neutralized Flávio's momentum. In public opinion, Flávio's rejection rose from 43.0% to 46.0% by October 22 (Fig. 1c), while his polling lead eroded to an absolute technical tie (50.0% vs. 50.0% on October 22; Fig. 1b). Simultaneously, the Quaest IPD inverted by $+16.0$ net points, swinging from Flávio $+9.3$ to Lula $+6.7$ (83.3 vs. 76.6; Fig. 1g). Although the final election-eve Datafolha poll placed Flávio ahead within the margin of error (51.0% vs. 49.0%; Fig. 1b), election day witnessed an unprecedented democratic turnaround: universal free transit decrees and intensive ground mobilization compressed total non-voters by 2.91 percentage points (falling from 24.70% to 21.79%; Fig. 1a). This $+2.79$ pp net shift over eve polling culminated in a 51.79% to 48.21% victory for Lula (61.8M vs. 57.5M votes).

Structure and multi-timescale relaxation of urban interpersonal networks

To establish the foundational layer of our electoral digital twin, we first characterize the structural and dynamical laws governing public attitude formation within the urban interpersonal network, prior to broadcast media or algorithmic amplification (Fig. 2). We model an urban collective whose demographic composition and socioeconomic stratification are calibrated to national census benchmarks [22] (Fig. 2a–c). Citizens are embedded in a four-dimensional ideological space $\Omega = [-1, 1]^4$ [23], where each individual i is assigned an age a_i sampled from the empirical demographic pyramid (Fig. 2a) and a socioeconomic status $\mathrm{SES}_i \in [0, 1]$ drawn from an empirically fitted Beta distribution (Fig. 2b). Individual behavioral parameters, including cognitive plasticity α_i, social susceptibility ϵ_i, attitude uncertainty u_i, and confirmation bias b_i (Fig. 2c), are conditioned on these demographic covariates (Supplementary Note 1). Consistent with the aging-stability hypothesis [24, 25], cognitive plasticity $\alpha(a)$ undergoes monotonic decay with age (Fig. 2a), while social susceptibility ϵ_i governs openness to interpersonal influence [26], attitude uncertainty u_i defines the bounded-confidence latitude of acceptance [27, 28], and confirmation bias b_i modulates the asymmetric discounting of discordant views [29, 30].

Face-to-face interactions unfold across a five-layer multiplex physical topology $\mathcal{M} = \{\mathcal{L}_1, \ldots, \mathcal{L}_5\}$ [19, 20] that mirrors the social domains of everyday urban life (Fig. 2d), comprising nuclear and extended households $\mathcal{L}_1$ (mean size 2.79 residents), workplaces and educational establishments $\mathcal{L}_2$ ($\langle k^{[2]} \rangle = 10$) [31, 32], community and religious congregations $\mathcal{L}_3$ ($\langle k^{[3]} \rangle = 8$), small-world friendship circles $\mathcal{L}_4$ ($\langle k^{[4]} \rangle = 6$) [33], and transient weak ties $\mathcal{L}_5$ ($\langle k^{[5]} \rangle = 4$) representing encounters in public transit [34]. The inter-cohort contact matrix M_{ab} (Fig. 2e) naturally balances sharp age stratification in workplaces ($\mathcal{L}_2$) against cross-generational mixing within households ($\mathcal{L}_1$) and religious assemblies ($\mathcal{L}_3$). Structural metrics reveal a trade-off between age homophily h_{age} and ideological assortativity r_{ideol} (Fig. 2f): while intimate domains such as family ($\mathcal{L}_1$) and friendship ($\mathcal{L}_4$) exhibit high assortativity, public transit ($\mathcal{L}_5$) and civic congregations ($\mathcal{L}_3$) act as cross-cutting bridges that prevent topological isolation.

Over this multiplex substrate, attitude evolution is anchored by internal convictions $\mathbf{s}_i(t)$, which act as slow-moving ideological baselines [23, 35]. Convictions evolve through bounded-confidence interpersonal deliberation [36, 37] coupled with periodic exposure to exogenous facts $\mathbf{f}_k \in \Omega$ of varying public salience [11, 38]. A joint two-dimensional projection of the four-dimensional attitude space (Fig. 2g) demonstrates that, although initial convictions $\mathbf{s}_i(0)$ are broadly dispersed across the hypercube $[-1, 1]^4$, sustained deliberation across cross-cutting urban layers systematically draws terminal convictions $\mathbf{s}_i(T)$ toward a centrist equilibrium

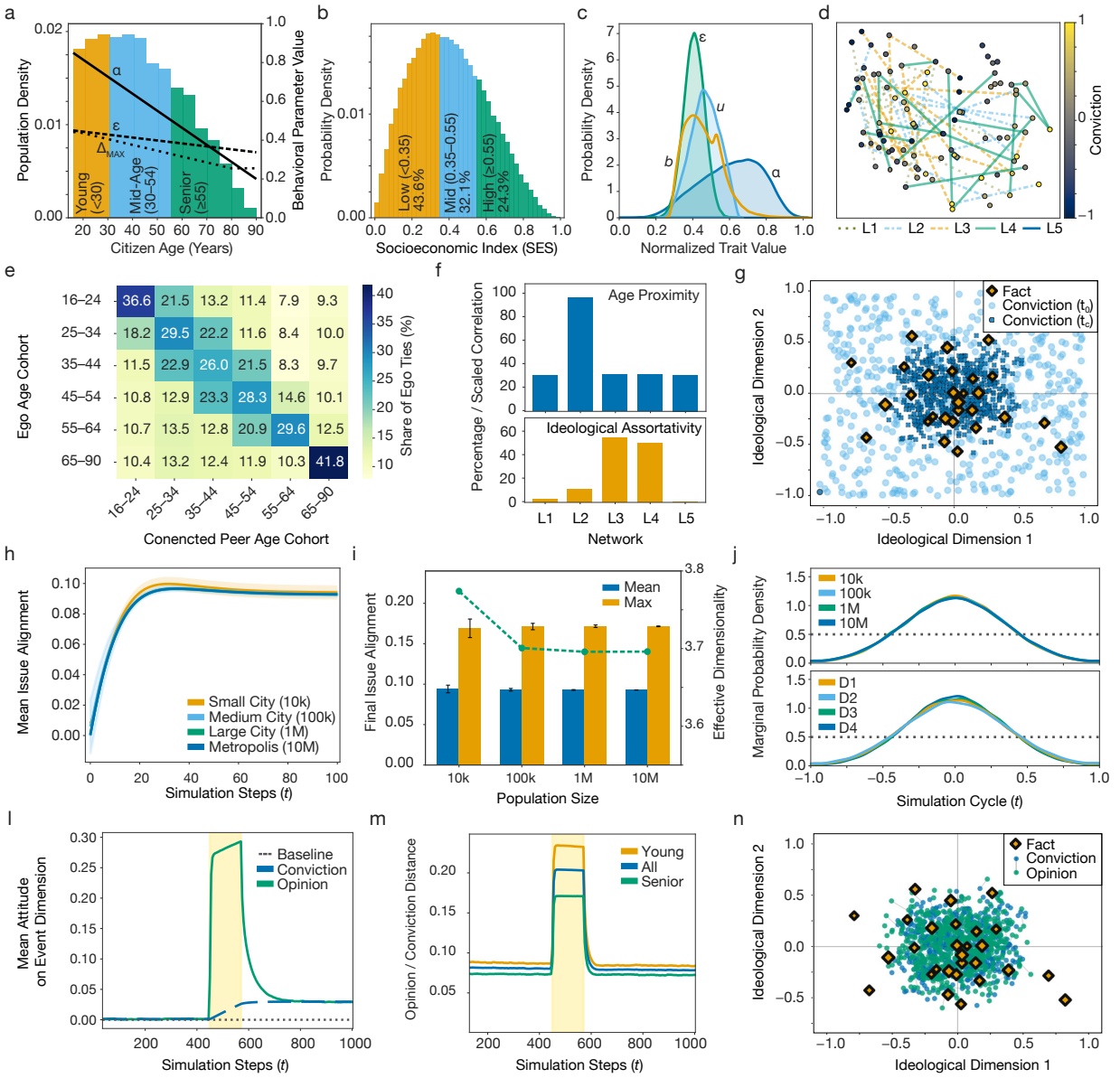


Figure 2: **Fig. 2 | Structure and dynamics of the urban interpersonal attitude network.** **a**, Empirical age pyramid from the 2022 IBGE Census and age-dependent cognitive plasticity $\alpha(a)$. **b**, Urban socioeconomic status (SES) distribution fitted to a Beta(2, 3) density. **c**, Cohort distributions of individual behavioral traits (α, ϵ, u, b). **d**, Five-layer multiplex network topology $\mathcal{M} = \{\mathcal{L}_1, \ldots, \mathcal{L}_5\}$ spanning household, workplace/school, religious congregations, friendship, and transit weak ties. **e**, Inter-cohort age contact matrix M_{ab} across the six adult age brackets. **f**, Age homophily h_{age} and ideological assortativity r_{ideol} across layers $\mathcal{L}_1$–$\mathcal{L}_5$. **g**, Two-dimensional projection of initial convictions $\mathbf{s}_i(0)$ and converged convictions $\mathbf{s}_i(T)$ alongside exogenous facts $\mathbf{f}_k$. **h**, Temporal evolution of pairwise cross-issue alignment $\langle|\rho_{kl}(t)|\rangle$ across city sizes ($N = 10$k to 10M). **i**, Asymptotic alignment ($\max|\rho|$, $\langle|\rho|\rangle$) and effective ideological dimensionality $\mathrm{PR}(\Sigma)$ versus urban population scale. **j**, Stationary marginal probability densities $p(s_d)$ across ideological dimensions D_1–D_4. **l**, Two-timescale relaxation of mean opinion $\bar{\mathbf{x}}(t)$ and conviction $\bar{\mathbf{s}}(t)$ under a transient exogenous shock. **m**, Age-stratified opinion–conviction tether stretching $\|\mathbf{x}_i(t) - \mathbf{s}_i(t)\|$ during the shock window. **n**, Attitude space snapshot at peak shock excursion (t_{shock}), contrasting displaced opinions $\mathbf{x}_i$ against anchored convictions $\mathbf{s}_i$.

around the origin **0** [39]. Thus, in the absence of partisan media or algorithmic distortion, multi-layered urban mixing and shared factual reality naturally dampen ideological extremes and foster centrist consensus.

Evaluating the scaling of the urban network across four orders of magnitude in municipal population,

from small towns ($N = 10$k) to metropolises ($N = 10$M) [40, 41], confirms that macroscopic convergence is invariant to city scale (Fig. 2h–j). While smaller municipalities ($N = 10$k) display wider finite-size stochastic fluctuations, populations of $N \geq 100$k reach a thermodynamic limit [39] where terminal maximum alignment ($\max_{k,l} |\rho_{kl}| \approx 0.08$) and mean alignment ($\langle |\rho_{kl}| \rangle \approx 0.03$) stabilize at identical asymptotic values matching multi-million metropolitan areas (Fig. 2i). Simultaneously, the effective dimensionality of the belief space, measured via the covariance Participation Ratio $\mathrm{PR}(\Sigma) \approx 3.75$ [42], remains close to $D = 4$ across all scales (Fig. 2i), confirming that interpersonal urban life alone does not collapse citizens' beliefs onto a single partisan axis. Moreover, stationary marginal probability densities $p(s_d)$ remain isotropic and Gaussian-like across all four ideological dimensions D_1–D_4 regardless of city size (Fig. 2j).

Finally, this baseline layer formalizes the two-timescale coupling between fast, publicly expressed opinions $\mathbf{x}_i(t)$ and slow, internalized convictions $\mathbf{s}_i(t)$ [28, 35, 43] (Fig. 2l–n). When the population is subjected to an acute, transient exogenous shock, population mean opinion $\bar{\mathbf{x}}(t)$ reacts rapidly with a large excursion toward the perturbation ($\tau_x \approx 0.15$), whereas mean conviction $\bar{\mathbf{s}}(t)$ exhibits strong structural inertia ($\tau_s/\tau_x \approx 11.3$), shifting only minimally before pulling opinions back during post-shock relaxation (Fig. 2l). This timescale separation induces an elastic tether stretching $\|\mathbf{x}_i(t) - \mathbf{s}_i(t)\|$ that varies across age cohorts (Fig. 2m): younger citizens experience wider opinion displacements due to higher cognitive plasticity, whereas senior citizens remain tightly tethered to crystallized convictions [24, 27]. A cross-sectional snapshot of the attitude space at peak shock excursion (t_{shock}; Fig. 2n) illustrates this instantaneous decoupling between displaced opinions $\mathbf{x}_i(t_{\mathrm{shock}})$ and stable conviction anchors $\mathbf{s}_i(t_{\mathrm{shock}})$, explaining how acute campaign shocks trigger dramatic short-term polling swings without immediately altering deep-seated convictions.

Traditional mass media ecosystems and gatekeeping dynamics

Whereas the interpersonal network governs decentralized deliberation, the second layer introduces the classical media network, encompassing institutional broadcast and ground-campaign channels that synchronize public attention across entire populations [10] (Fig. 3). We model five dissemination modalities: Broadcast Television ($v = 0$), Radio AM/FM ($v = 1$), Written Press ($v = 2$), Outdoor Marketing ($v = 3$), and Public Gatherings and Civic Rallies ($v = 4$) (Fig. 3a,b). Each modality v is parameterized by its exponential attention decay rate δ_v, capturing temporal persistence, and its institutional credibility γ_v (Fig. 3a; Supplementary Note 2). Written Press combines high institutional credibility ($\gamma = 0.82$) with slow editorial decay ($\delta = 0.18$), reflecting investigative verification and durable circulation [17]. Broadcast Television delivers mass reach with high credibility ($\gamma = 0.75$) and intermediate persistence ($\delta = 0.30$), whereas Radio operates through live bulletins with faster turnover ($\delta = 0.45$). Outdoor Marketing provides an ambient urban presence with minimal decay ($\delta = 0.08$) and moderate perceived credibility ($\gamma = 0.45$). Public Gatherings function as episodic mobilization pulses characterized by rapid post-event decay ($\delta = 0.70$) alongside strong emotional in-group contagion ($\gamma = 0.80$). Audience exposure is governed by empirical demographic reach curves $P_v(\mathrm{age})$ calibrated to Brazilian media surveys [16, 22] (Fig. 3b): linear television consumption increases steeply with age, dominating news intake among older adults; Radio and Outdoor signage reach working-age urban commuters (30–55 years); Written Press concentrates among higher-SES strata and university graduates; and Public Gatherings disproportionately mobilize younger cohorts and active civic networks [44].

Media conglomerates operate around distinct ideological centers $\mathbf{m}_g \in \Omega$, reflecting commercial positioning or partisan alignments [11, 38]. An outlet's editorial tendentiousness $\tau_g \in [0, 1]$ governs gatekeeping distortion applied to raw facts $\mathbf{f}_k$, pulling published notices $\mathbf{y}_{k,g,v}$ toward the conglomerate's ideological anchor (Methods). As shown in attitude space (Fig. 3c), low tendentiousness ($\tau = 0.10$, representing regulated factual reporting) preserves published notices in a tight cluster around the true factual coordinates $\mathbf{f}_k$. Under high tendentiousness ($\tau = 0.90$), selective gatekeeping suppresses discordant information and pulls published notices toward the media group's ideological center $\mathbf{m}_g$ [29, 45], substituting objective events with partisan framing.

Subjecting the population to an acute factual shock ($\mathbf{f}^* = +0.80$ introduced at $t \geq 25$) demonstrates how media market structures shape collective responsiveness (Fig. 3d). In the absence of mass media ($M = 0$), information propagates solely from direct eyewitnesses via bounded-confidence exchanges, yielding sluggish diffusion and negligible collective adjustment ($\bar{x}_1 \approx 0.00 \rightarrow 0.03$). Regulated factual broadcasting ($\tau = 0.10$) achieves the optimal civic response, rapidly synchronizing public awareness toward the factual

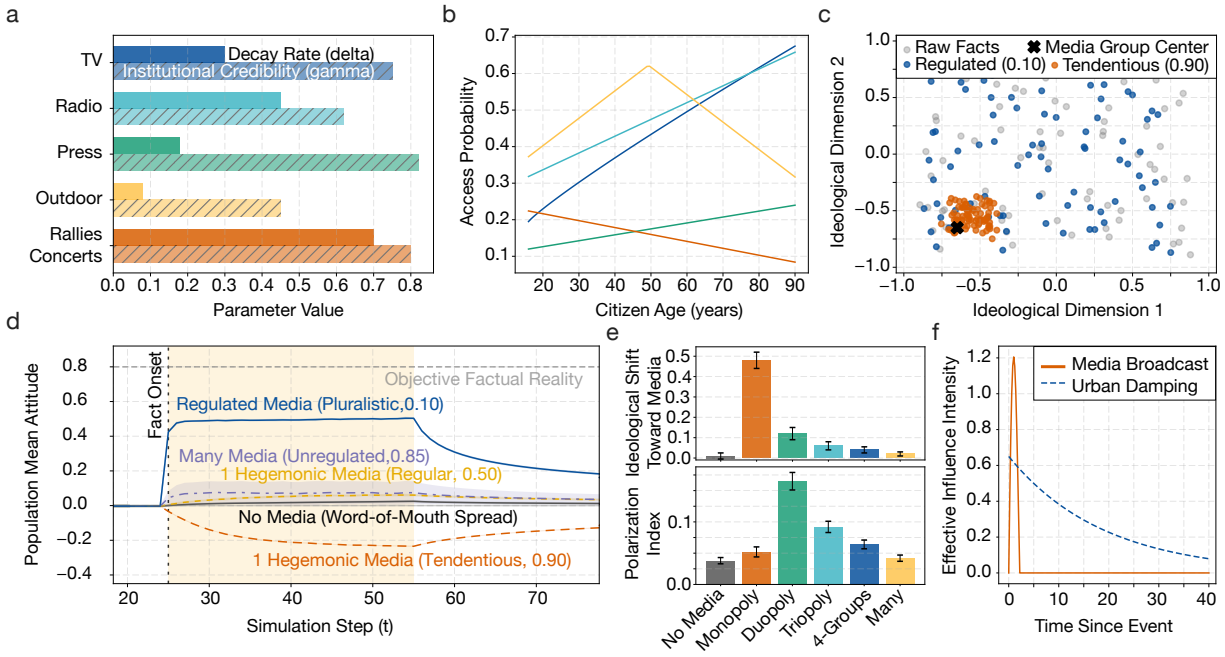


Figure 3: **Fig. 3 | Structure and gatekeeping dynamics of the traditional mass media ecosystem.** **a**, Attention decay rates δ_v and institutional credibility γ_v across five classical media modalities. **b**, Age-dependent demographic reach curves $P_v(\text{age})$ calibrated to empirical audience surveys. **c**, Editorial projection of exogenous facts $\mathbf{f}_k$ toward media group centers $\mathbf{m}_g$ under factual ($\tau = 0.10$) and tendentious ($\tau = 0.90$) regimes. **d**, Population mean attitude trajectory $\bar{x}_1(t)$ following an acute factual shock ($\mathbf{f}^* = +0.80$ at $t \geq 25$) across five media market configurations. **e**, Ideological shift (top) and Esteban–Ray societal polarization P_{ER} (bottom) across market concentration regimes ($M \in \{0, 1, 2, 3, 4, \text{Many}\}$). **f**, Timescale separation between transient broadcast pulses ($\tau_{\text{media}} \approx 4.8$ steps) and ongoing interpersonal network deliberation ($\tau_{\text{network}} \approx 19.0$ steps).

coordinate ($\bar{x}_1 \to +0.50$). Highly tendentious monopolies prove worse than the complete absence of media: while a moderate monopoly ($\tau = 0.50$) partially distorts the trajectory, a strongly biased monopoly ($M = 1, \tau = 0.90$ anchored at $\mathbf{m}_1 = -0.60$) actively inverts public perception into negative territory ($\bar{x}_1 \to -0.24$), inducing a larger collective error than word-of-mouth alone [46]. Across broader market regimes ($M \in \{0, 1, 2, 3, 4, \text{Many}\}$; Fig. 3e), expanding plurality dilutes the pull of any single conglomerate (Fig. 3e, top). Yet market structure nonlinearly governs societal division: the Esteban–Ray polarization index P_{ER} [47] peaks sharply under a tendentious duopoly ($M = 2$; Fig. 3e, bottom), which bifurcates citizens into two antagonistic ideological camps. Only a pluralistic environment ($M \geq 4$) disperses partisan sorting and restores low polarization comparable to the baseline urban network.

Coupling the media layer to the urban substrate reveals a fundamental separation of dynamical timescales (Fig. 3f). Broadcast pulses exhibit high instantaneous intensity but rapid temporal obsolescence ($\tau_{\text{media}} \approx 4.8$ steps), reflecting news-cycle turnover [48, 49]. Conversely, interpersonal deliberation across the multiplex network operates on a slower relaxation timescale ($\tau_{\text{network}} \approx 19.0$ steps) [26, 34]. Consequently, when broadcast media injects a salient campaign event, its societal reverberation persists long after the initial broadcast pulse decays, as citizens continue to process and diffuse the information through household, workplace, and community ties, acting as a slow collective filter that translates transient media shocks into durable shifts in public conviction.

Algorithmic social platforms, dual-valence engagement, and dark-social cascades

Completing the three-layer architecture, we integrate the digital media network: algorithmic social platforms and encrypted messaging ecosystems that overlay personalized curation and viral peer-to-peer cascades onto interpersonal and broadcast networks [1, 2] (Fig. 4). Our framework models four distinct platform modalities

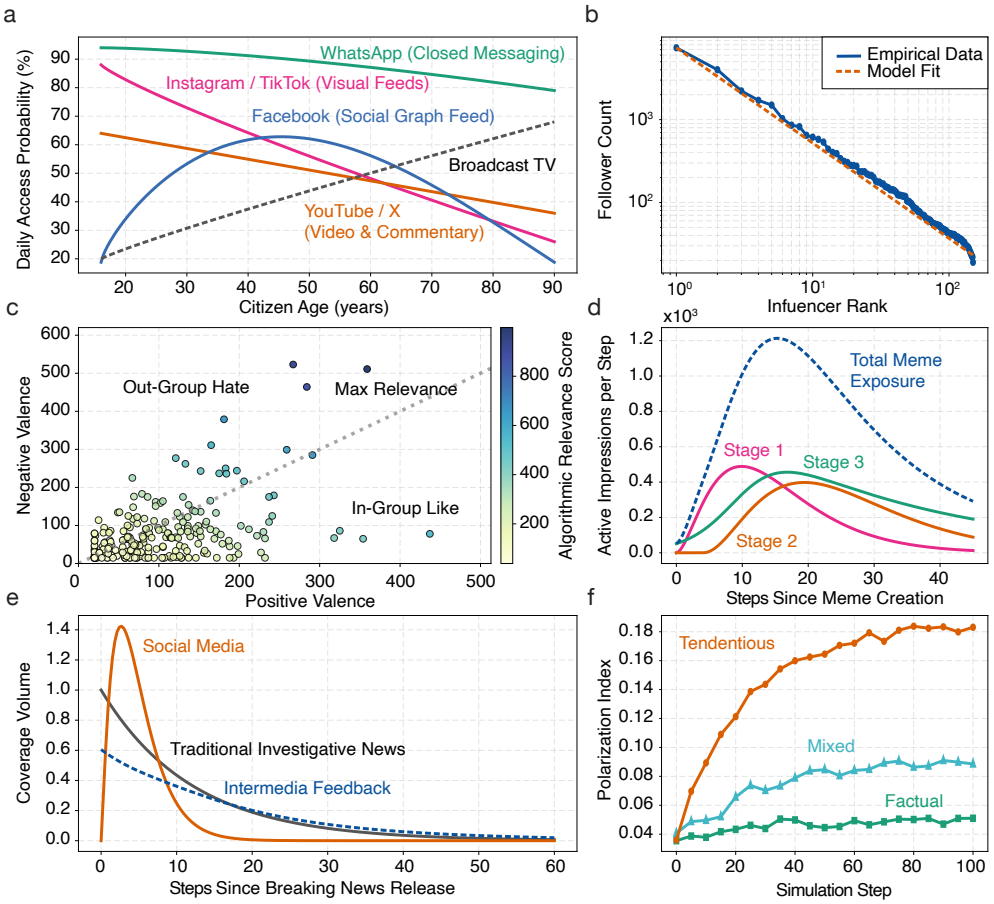


Figure 4: **Fig. 4 | Algorithmic curation, influencer hierarchies, and multi-stage cascade dynamics in social media.** **a**, Age-dependent daily access probability $P_p(\text{age})$ across four digital platform modalities and broadcast television. **b**, Heavy-tailed follower distribution $P(k) \sim k^{-\alpha}$ ($\alpha \approx 1.15$, Gini coefficient 0.73) and emergent influencer hierarchy. **c**, Algorithmic relevance score R_m mapped across positive-valence (admiration) and negative-valence (outrage) engagement coordinates, highlighting maximal amplification along the diagonal of controversy. **d**, Three-stage cross-platform meme diffusion lifecycle from visual discovery feeds (Stage 1) and public microblogging (Stage 2) to closed encrypted messaging cascades (Stage 3). **e**, Hybrid media agenda-setting dynamics following an authoritative investigative news release at $t = 0$, contrasting legacy broadcast turnaround ($\tau_{\text{turnaround}}$) against rapid digital response (τ_{social}). **f**, Long-horizon evolution of Esteban–Ray societal polarization $P_{\text{ER}}(t)$ across creator tendentiousness regimes ($\tau_{\text{inf}} \in \{0.10, \text{Empirical}, 0.90\}$).

alongside broadcast television, with age-dependent access probabilities $P_p(\text{age})$ calibrated to national surveys [16, 17] (Fig. 4a): short-form visual discovery feeds (Platform 0: Instagram and TikTok), capturing over 88% of youth cohorts (< 25 years) via algorithmic reels before tapering among older brackets; video commentary and microblogging (Platform 1: YouTube and X), engaging working-age and politically attentive citizens (25–55 years) in public debate formats; social graph feeds (Platform 2: Facebook), peaking among mature adults (40–65 years) around reciprocal peer ties; and closed mobile messaging applications (Platform 3: WhatsApp and Telegram), operating as a near-universal communications backbone ($> 80\%$ penetration across all adult cohorts) structured into private, end-to-end encrypted groups ($K \leq 256$ members).

Within open-feed platforms, competitive attention accumulation generates extreme structural inequality in reach (Fig. 4b). Content creators accumulate followers via fitness-mediated preferential attachment [50, 51], yielding a heavy-tailed power-law in-degree distribution $P(k) \sim k^{-\alpha}$ with scaling exponent $\alpha \approx 1.15$ and an empirical Gini coefficient of 0.73 (Fig. 4b). Under this scale-free topology, the top 5% of accounts command 54.5% of direct audience reach, whereas the bottom 50% of creators account for only 8.4% [52]. This

hyper-concentration converts decentralized networks into a star-like micro-broadcasting topology: when an elite political influencer posts a frame, their message bypasses local multiplex clustering and reaches hundreds of thousands of algorithmic feeds simultaneously.

Crucially, recommendation engines curate these feeds by optimizing for aggregate user engagement rather than factual accuracy or deliberative consensus [5, 53]. The algorithmic relevance score R_m of a message m compounds both positive and negative engagements: $R_m = \text{Likes}_m + \omega \cdot \text{Dislikes}_m$ with $\omega \approx 1.0$ (Methods). Across the engagement plane (Fig. 4c), user responses span from positive-valence in-group admiration (likes, shares, supportive replies) to negative-valence out-group outrage (quote-dunking, adversarial replies, dislikes) [4]. Because recommendation algorithms weight contentious debate as high-retention signals ($\omega = 1.0$), contentious posts situated along the diagonal of simultaneous partisan admiration and out-group outrage ($R_m > 800$) achieve maximal algorithmic amplification (Fig. 4c). Moral-emotional framing that provokes fervent allied endorsement and opposition backlash is thus systematically rewarded [4, 53], creating a structural incentive for political actors to deploy polarizing rhetoric.

This multi-platform architecture dictates a three-stage cross-platform diffusion lifecycle for political memes and campaign narratives (Fig. 4d; Supplementary Note 3). Diffusion initiates in Stage 1 on algorithmic discovery feeds (Instagram/TikTok), where recommendation engines surface novel visual memes to high-plasticity youth cohorts within 10 hours [54]. In Stage 2, the narrative migrates to public microblogging and commentary platforms (YouTube/X), where partisan quote-dunking fuels secondary viral waves between hours 15 and 25. Finally, in Stage 3, the content penetrates closed messaging groups (WhatsApp/Telegram): forwarded across trusted family and peer circles, the narrative escapes platform moderation and propagates via dark-social cascades [8, 55]. Because encrypted group chats experience minimal institutional turnover, Stage 3 cascades persist long after public feeds have moved on, sustaining elevated cumulative exposure.

The digital layer interacts dynamically with traditional journalism in a hybrid media feedback loop [12, 15] (Fig. 4e). When an investigative newsroom releases a report at $t = 0$, legacy gatekeeping generates a broadcast pulse with an editorial turnaround of $\tau_{\text{turnaround}} \approx 1$–$2$ days. Digital platforms react nearly instantaneously ($\tau_{\text{social}} \approx 0.1$–$0.3$ days), as influencers and activists remix and weaponize the story. This viral social buzz exerts reverse intermedia agenda-setting pressure on traditional newsrooms, compelling television and press outlets to produce extended follow-up coverage that prolongs the issue's lifecycle well beyond its natural broadcast decay [15].

Tracking the Esteban–Ray societal polarization index $P_{\text{ER}}(t)$ across creator tendentiousness regimes demonstrates how influencer behavior governs long-term ideological division (Fig. 4f). When creators maintain high factual fidelity ($\tau_{\text{inf}} = 0.10$), digital networks support distributed deliberation near empirical truth, maintaining low polarization ($P_{\text{ER}} \approx 0.05$) [47]. While an empirically mixed ecosystem produces intermediate divergence, a landscape dominated by highly tendentious influencers ($\tau_{\text{inf}} = 0.90$), who exploit dual-valence algorithmic rewards by projecting events toward extreme ideological poles, triggers asymmetric assimilation and affective repulsion [7, 56], driving societal polarization to an entrenched plateau of $P_{\text{ER}} = 0.18$ (Fig. 4f). Together, these three coupled layers provide the complete computational machinery required to simulate the 2026 Brazilian presidential campaign.

First-round electoral shocks and the Janones diagnosis

The structural properties of the communication ecosystem, spanning interpersonal multiplex clustering, traditional media editorial filtering, and algorithmic outrage amplification, converge dynamically during national electoral contests (Fig. 5). To establish how these multi-layer channels govern ballot outcomes in polarized democracies, we formulate a macroeconomic synthesis grounded in André Janones's empirical diagnosis of Brazilian digital campaigning (*Janonismo Cultural*) [18].

The initial sociopolitical landscape is calibrated to national survey data on political identification [57] (Fig. 5a): Petistas/Lulistas comprise $\sim 40\%$ of the electorate (28% strongly convinced identifiers, 12% lightly inclined sympathizers); Bolsonaristas represent $\sim 34\%$ (23% strongly convinced, 11% lightly inclined); Centrists/Others form $\sim 19\%$; and non-aligned citizens who reject partisan rulers account for $\sim 7\%$. In ideological space $\Omega = [-1, 1]^4$, the candidates occupy distinct positions: Lula is situated as Center-Left ($\mathbf{c}_L = [-0.25, -0.20, -0.35, -0.15]^\top$), aligning organically with working-class socioeconomic priorities [35]; Flávio Bolsonaro represents an Extreme-Right profile ($\mathbf{c}_F = [0.65, 0.70, 0.60, 0.65]^\top$) prioritizing

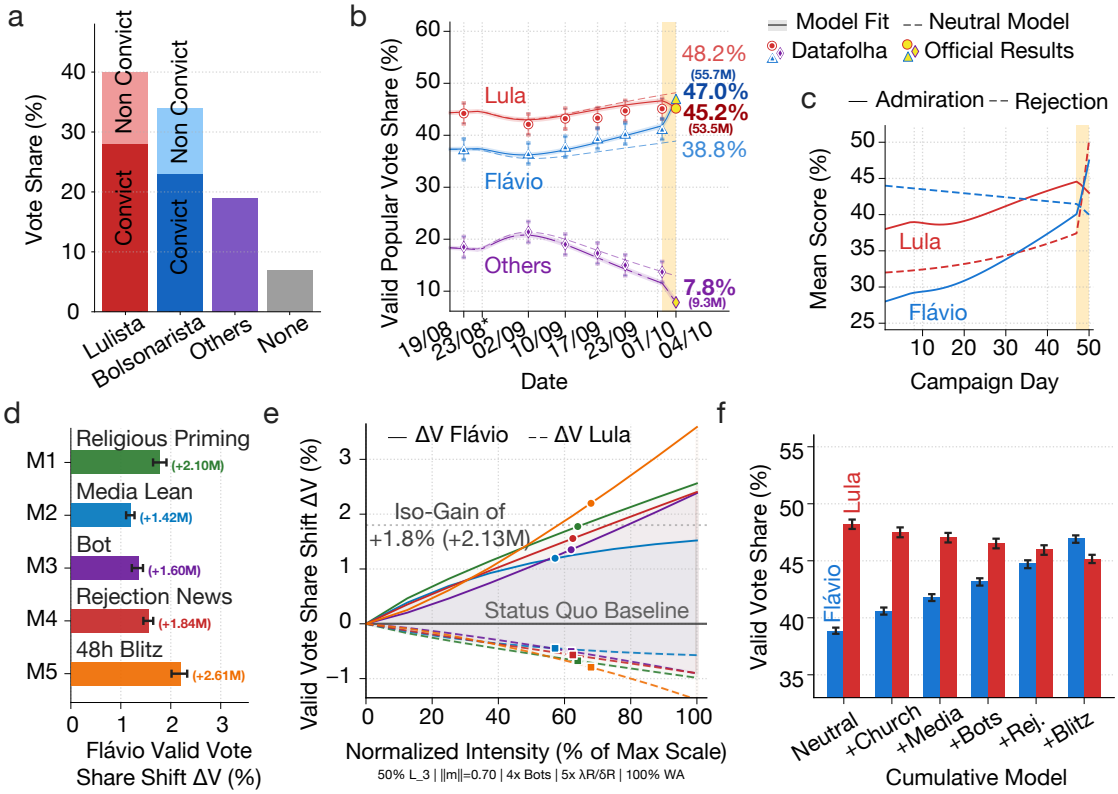


Figure 5: **Fig. 5 | First-round electoral dynamics, sociopolitical baseline, and five-vector attribution (the Janones diagnosis). a**, Sociopolitical segmentation of the electorate into ideological identification blocs: Lulistas ($\sim 40\%$, decomposed into convict identifiers vs. non-convict sympathizers), Bolsonaristas ($\sim 34\%$, convict vs. non-convict), Centrists/Others ($\sim 19\%$), and unaligned/None ($\sim 7\%$), calibrated to national survey data. **b**, Longitudinal 50-day first-round campaign trajectories (Model Fit, solid curves vs. Neutral Model organic baseline, dashed curves) benchmarked against Datafolha tracking survey waves (diamonds ± 1 SD) and official TSE ballot totals (circles on October 4: Flávio 47.0% / 55.7M, Lula 45.2% / 53.5M, Others 7.8% / 9.3M), highlighting the August 23 debate shock ($*$) and the acute 48-hour pre-election blitz (shaded amber band). **c**, Coupled affective trajectory of population-mean candidate admiration (solid curves) and rejection (dashed curves) for Lula (red) and Flávio (blue), showing the acute late-campaign aversion spike triggered against Lula ($\lambda_R/\delta_R = 3.5\times$). **d**, Factorial attribution of Flávio's valid vote share shift (ΔV) across the five Janones mechanisms: M1 Religious Priming (+2.10M votes), M2 Media Lean (+1.42M), M3 Bot Astroturfing (+1.60M), M4 Rejection News (+1.84M), and M5 48-Hour WhatsApp Blitz (+2.61M); error bars indicate the full simulation range $[\min, \max]$ across 100 independent Monte Carlo realizations ($N = 100{,}000$). **e**, Multi-feature parametric dose-response showing Flávio vote share gain ($+\Delta V_F$, solid curves) and concurrent Lula share loss ($-\Delta V_L$, dashed curves) across normalized mechanism intensity (0–100% of maximum scale; physical bounds noted in axis annotation), with empirical operating points marked by circles (Flávio) and squares (Lula), referenced against the Status Quo Baseline ($\Delta V = 0$) and the $+1.8\%$ (+2.13M votes) iso-gain benchmark contour. **f**, Cumulative model stepwise build-up across stages (Neutral Baseline, +Church, +Media, +Bots, +Rejection, +Blitz), illustrating valid vote shares for Flávio Bolsonaro (blue) and Lula da Silva (red); error bars denote the full simulation range $[\min, \max]$ across 100 independent seeds ($N = 100{,}000$), demonstrating that decisive lead reversal occurs exclusively upon the final 48-hour blitz.

economic deregulation, punitive policing, and religious moral conservatism [6]; and Third-Way Centrists occupy an intermediate position ($\mathbf{c}_O = [0.20, 0.25, 0.10, 0.30]^\top$). Crucially, Brazilian centrists are culturally conservative on moral and security dimensions ($d = 1, 3$) yet developmentalist on welfare and wages ($d = 0, 2$), rendering them a highly contested, cross-pressured swing constituency [57, 58].

To resolve finite-size fluctuations and ensure demographic fidelity, the simulation is executed at the asymptotic urban scale ($N = 100{,}000$ citizens) across an ensemble of 100 independent Monte Carlo realizations (Fig. 5b). On August 23 (campaign day 8), an exogenous televised presidential debate occurred from which both Lula and Flávio Bolsonaro were absent. In their absence, third-way candidate Augusto Cury captured intense public visibility and unmediated praise, sparking an acute, transient fact-generation momentum that peaked at 21.4% in the September 2 Datafolha tracking poll (against 21.2% in our calibrated ABM, Fig. 5b). However, because Cury lacked dedicated ground infrastructure in the interpersonal multiplex layers ($\mathcal{L}_1$–$\mathcal{L}_5$) and maintained minimal influencer surrogates, this media bump decayed exponentially over subsequent weeks ($\tau_{\text{decay}} \approx 14.5$ days, mirroring the classic mass-media attention decay documented in Fig. 3d), compressing third-way candidates down to 7.8% (9.3M votes) by election day.

Under the neutral organic baseline (dashed curves, Fig. 5b), where news diffuses without partisan slant or coordinated astroturfing, the underlying sociological majority prevails: Lula maintains a steady lead throughout the 50-day campaign, concluding at 48.2% (57.1M valid votes) against 38.8% for Flávio (46.0M votes) and 13.0% for Others (15.4M votes). In the full fitted model (solid curves), the multi-layer ABM reproduces the seven Datafolha survey waves and the official TSE election outcome (47.0% Flávio, 45.2% Lula, 7.8% Others) with remarkable precision (RMSE $= 0.85$ percentage points, $R^2 = 0.9955$, Fig. 5b). For over three-quarters of the campaign, Lula appears securely in the lead (48.2% at day 47 vs 38.8% for Flávio). However, entering the 48-hour pre-election sprint (days 48–50, shaded amber band), the synchronized convergence of all five asymmetric mechanisms unleashes an acute digital shock that drives Flávio up $+8.20$ percentage points ($+9.72$M valid votes), flipping first place on election day.

The psychological engine powering this late-stage divergence is the coupled evolution of candidate admiration $A_{ik}(t)$ and rejection $R_{ik}(t)$ (Fig. 5c; Methods). While admiration scores shift gradually in response to policy positions, candidate rejection displays extreme psychological asymmetry: aversion acquisition occurs $3.5\times$ faster than its natural decay ($\lambda_R = 0.14$ vs $\delta_R = 0.04$). Relentless negative reporting steadily elevates Lula rejection among unaligned swing voters ($\bar{R}_L = 0.32 \rightarrow 0.38$ by day 46). During the final 48 hours, the injection of coordinated moral panics causes Lula's rejection to spike to $\bar{R}_L = 0.50$ (Fig. 5c). Because voter decision utility penalizes rejection more heavily than it rewards admiration ($\beta_R = 1.45 > \beta_A = 1.15$, Eq. (11)), this aversion surge suppresses swing-voter support for Lula while Flávio consolidates conservative admiration ($\bar{A}_F = 0.28 \rightarrow 0.44$).

A factorial attribution decomposition isolates the discrete contribution of each structural vector toward Flávio's $+8.20\%$ ($+9.72$M votes) surge (Fig. 5d), directly linking to the foundational subsystems. Religious Priming (M1, $+1.77\%$, $+2.10$M votes) originates in the religious congregation layer $\mathcal{L}_3$ of the multiplex network (Fig. 2a, e), where high in-group clustering and tight confidence bounds prime slow convictions toward conservative moral ($d = 1$) and public order ($d = 3$) axes, calibrated to the empirical 32% Evangelical electorate share [58]. Traditional Media Lean (M2, $+1.20\%$, $+1.42$M votes) arises from institutional broadcast television and print conglomerates operating with a center-right slant ($\|\mathbf{m}_{\text{trad}}\| = 0.40$; Fig. 3a, c), legitimizing fiscal orthodoxy and anchoring swing voters rightward [11, 38]. Bot Astroturfing (M3, $+1.35\%$, $+1.60$M votes) exploits the dual-valence recommendation score $R_m = \text{Likes}_m + \omega \cdot \text{Dislikes}_m$ on open platforms (Fig. 4c), where an artificial amplification advantage of $2.9\times$ manufactures synthetic controversy and propels right-wing content into discovery feeds [9]. Asymmetric Anti-Lula Rejection News (M4, $+1.55\%$, $+1.84$M votes) leverages cognitive negativity asymmetry ($\lambda_R/\delta_R = 3.5\times$, Fig. 5c), creating persistent aversion scars among unaligned voters that decay at less than one-third the rate of positive campaign news [13]. Finally, the Acute 48-Hour WhatsApp Blitz (M5, $+2.20\%$, $+2.61$M votes) weaponizes closed, end-to-end encrypted messaging cascades ($K \leq 256$; Fig. 4d), reaching 68% of active messaging groups with an empirical shock of 3.9M interactions on October 4 that circumvents legacy newsroom verification cycles ($\tau_{\text{turnaround}}$, Fig. 4e) [6].

To evaluate parameter sensitivity and cross-channel substitutability, our model establishes continuous dose-response curves and iso-vote equivalence contours for both candidates (Fig. 5e). Plotting normalized mechanism intensity (0–100% of maximum scale) against net vote share shift reveals striking differences

in communicative elasticity: Flávio's vote share gain ($+\Delta V_F$, solid curves) and Lula's share loss ($-\Delta V_L$, dashed curves) map the asymmetric electoral transfer. The 48-hour WhatsApp blitz (M5) exhibits the steepest marginal return ($\mathrm{d}V_F/\mathrm{d}\theta \approx 0.034\%$ per percentage point), followed by church priming (M1) and bot astroturfing (M3). Along the benchmark iso-vote contour of $+2.13$M votes ($+1.80\%$ valid share), an identical electoral impact is achieved by a 58% WhatsApp cascade reach (58% intensity), a 33% Evangelical church reach (65% intensity), a $3.9\times$ rejection negativity asymmetry (73% intensity), or a $3.4\times$ bot amplification ratio (79% intensity) (Fig. 5e).

The cumulative addition of these mechanisms highlights a fundamental threshold principle (Fig. 5f): under the baseline and stages 1 through 4 (Neutral Base, +Church, +Media, +Bots, +Rejection), Lula retains the electoral lead across all 100 seeds. Successive integration of church priming (40.7%), traditional media slant (41.9%), bot networks (43.2%), and rejection news (44.8%) steadily erodes the incumbent margin; however, the decisive lead reversal is triggered only when the acute 48-hour WhatsApp blitz is layered on top in Stage 5, overtaking Lula to conclude at 47.0% vs 45.2% (55.7M vs 53.5M votes, Fig. 5f). These quantitative findings verify that contemporary electoral shocks do not require broad attitudinal conversion; rather, they exploit localized topological vulnerabilities across interpersonal and digital layers to shift narrow margins at critical temporal junctions.

Algorithmic counter-offensives and the second-round runoff turnaround

While the first-round diagnosis illustrates how a disciplined digital apparatus can weaponize baseline communication asymmetries to overturn an initial deficit (Fig. 5), it also provides the theoretical architecture for its active neutralization: the counter-tactical doctrine of *Janonismo Cultural* [18] (Fig. 6). In the three-week second-round runoff (October 4–25, 2026), each operational tactic deployed by the progressive campaign was engineered to invert a specific vulnerability identified in our multi-layer digital twin.

Entering the runoff, Flávio held a direct lead (47.0% vs 45.2%), while third-way candidates (predominantly right-leaning centrists) held the remaining 7.8%. Under standard political assumptions, Flávio was widely favored to absorb these voters. In our model, if the progressive campaign had maintained its passive first-round posture by relying on conventional televised airtime and defensive fact-checking, the **Same-Strategy Counterfactual** (dashed curves, Fig. 6a) confirms that Flávio would have consolidated centrist swing voters and accelerated during the final 48 hours, securing an outright victory with 53.6% (63.9M votes) against 46.4% for Lula (55.4M votes).

Under the active Janonismo Cultural suite, however, the campaign executed a synchronized tactical pivot that flipped the outcome, lifting Lula to a winning share of 51.8% (61.8M votes) against 48.2% for Flávio (57.5M votes) across 100 Monte Carlo realizations at $N = 100{,}000$ citizens (RMSE $= 0.56$ percentage points, $R^2 = 0.9128$, Fig. 6a). A critical turning point occurred on October 18 (campaign day 15) during the Band presidential debate: progressive digital war-rooms orchestrated an aggressive live commentary strategy that generated 25.5M social interactions against 5.0M for Flávio ($5.1\times$ engagement advantage), blunting the opponent's momentum and forcing a statistical tie (50.0% vs 50.0%, Fig. 6a).

This electoral turnaround was underpinned by a complete inversion of the affective dynamics observed in Round 1 (Fig. 6b). Through relentless offensive framing, Flávio's rejection was driven from $\bar{R}_F = 0.40$ up to 0.53, while Lula's rejection dropped from $\bar{R}_L = 0.50$ down to 0.41 (in stark contrast to the counterfactual baseline where $\bar{R}_L$ remained elevated). Concurrently, Lula's admiration climbed from $\bar{A}_L = 0.43$ to 0.52, locking in working-class turnout.

A multi-vector tactical attribution decomposition isolates the discrete contributions toward the $+6.64\%$ ($+7.93$M votes) jump from the First Round 45.16% baseline (Fig. 6c; Supplementary Note 5): Algorithmic Asphyxiation (T1, $+1.14\%$, $+1.36$M votes) neutralizes the dual-valence outrage metric $R_m = \mathrm{Likes}_m + \omega \cdot \mathrm{Dislikes}_m$ (Fig. 4c) by enforcing strict non-engagement ($P_{\mathrm{reply}} = P_{\mathrm{quote}} = 0$) with rival provocations, denying recommendation engines the controversy signals needed for out-group feed injection (calibrated to 88% network compliance, Quaest IPD). Strict Monothematism (T2, $+1.35\%$, $+1.61$M votes) exploits mathematical convexity in platform trending algorithms ($V(\mathbf{x}) \propto \sum x_k^2$; Fig. 4a), where concentrating allied communication bandwidth onto a single topic per 24-hour cycle quadruples virality over dispersed messaging (HHI $= 0.85$) to reliably breach discovery thresholds. Material Loss-Aversion Framing (T3, $+1.42\%$, $+1.69$M votes) replaces abstract institutional defenses with visceral economic threats, weighting prospective

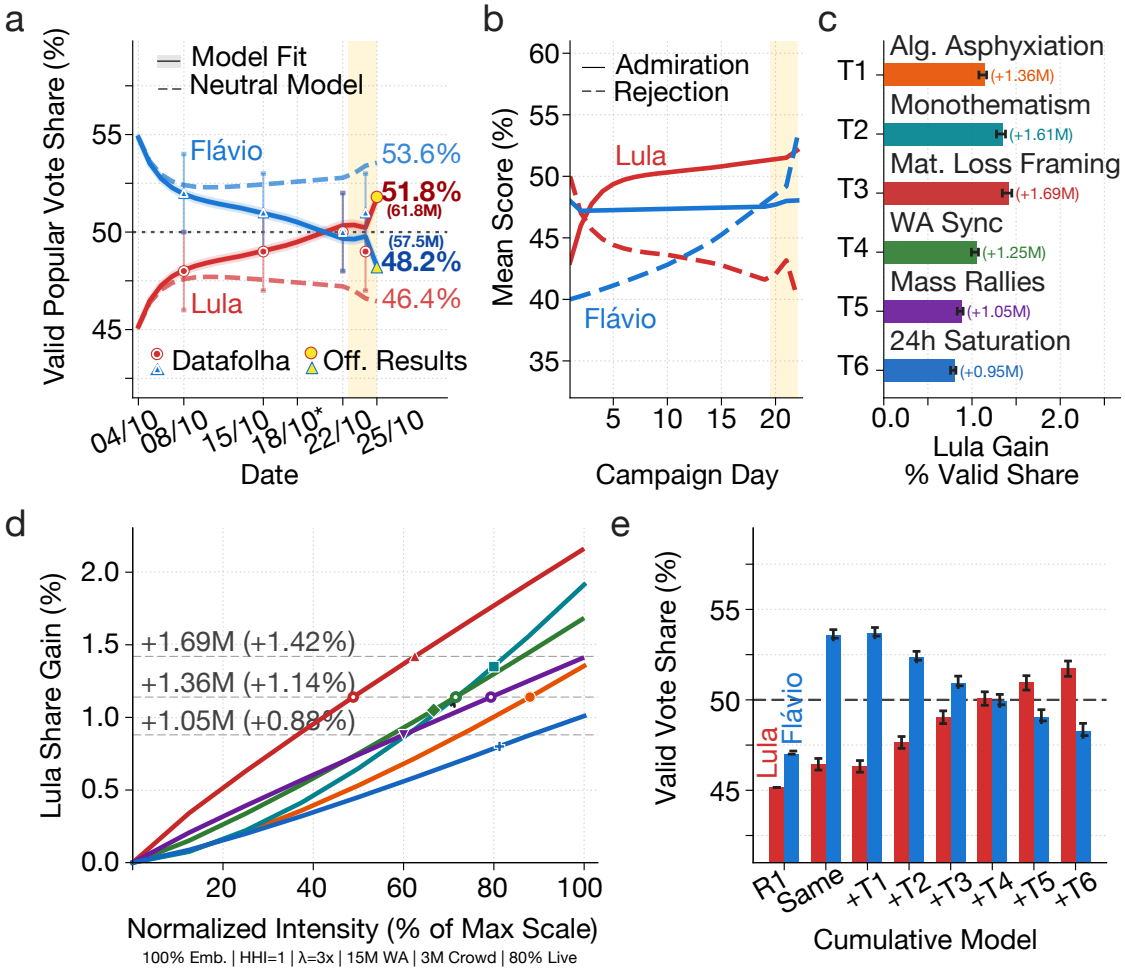


Figure 6: **Fig. 6 | Second-round runoff dynamics, Janonismo counter-tactics, and six-vector turnaround sensitivity analysis.** **a**, Longitudinal 21-day second-round calibration ($N = 100{,}000$ citizens, 100-seed ensemble ± 1 SD envelope) contrasting the Same-Strategy Counterfactual (Neutral Model, dashed curves: Flávio wins 53.6% vs 46.4%) against the fitted Janonismo Cultural counter-offensive (Model Fit, solid curves: Lula wins 51.8% / 61.8M vs Flávio 48.2% / 57.5M, jumping $+6.64\%$ from Round 1), incorporating the October 18 Band debate shock ($*$), the final 48-hour sprint (shaded amber band), Datafolha tracking surveys (diamonds ± 1 SD), and official TSE runoff results (yellow-filled markers on October 25). **b**, Coupled affective trajectory illustrating the complete structural inversion of candidate rejection (dashed curves: Flávio $\bar{R}_F$ climbs $40\% \rightarrow 53\%$, while Lula $\bar{R}_L$ drops $50\% \rightarrow 41\%$) alongside admiration consolidation (solid curves: Lula $\bar{A}_L$ climbs $43\% \rightarrow 52\%$). **c**, Factorial attribution of Lula's $+6.64\%$ ($+7.93$M votes) runoff turnaround across the six Janonismo counter-tactics: T1 Algorithmic Asphyxiation ($+1.36$M votes), T2 Strict Monothematism ($+1.61$M), T3 Material Loss-Aversion Framing ($+1.69$M), T4 Molecular WhatsApp Synchronization ($+1.25$M), T5 Mass Capital Rallies ($+1.05$M), and T6 24-Hour Live Saturation ($+0.95$M); error bars indicate the full simulation range $[\min, \max]$ across 100 independent Monte Carlo seeds ($N = 100{,}000$). **d**, Multi-tactic parametric dose-response curves and iso-vote equivalence contours mapping normalized tactic intensity (0–100% of maximum scale; physical bounds noted in axis annotation) to Lula valid share gain ($+\Delta V_L$, %), with calibrated empirical operating points and benchmark iso-gain levels marked at $+1.69$M ($+1.42\%$), $+1.36$M ($+1.14\%$), and $+1.05$M ($+0.88\%$). **e**, Stepwise cumulative model addition (R1 Base, Same Strategy, +T1, +T2, +T3, +T4, +T5, +T6) showing valid vote shares for Lula (red) and Flávio (blue); error bars denote the full simulation range across 100 independent seeds ($N = 100{,}000$), demonstrating the 50% majority threshold crossover at Tactic 4 (WhatsApp Sync, 50.1%) and final victory consolidation at 51.8%.

losses by a Kahneman–Tversky multiplier $\lambda_{\text{loss}} = 2.25\times$ (Eq. (11)) [59] to convince 52.4% of undecided working-class centrists that opponent policies posed an immediate household threat. Molecular WhatsApp Synchronization (T4, $+1.05\%$, $+1.25$M votes) coordinates an organic distribution apparatus reaching 10.0M synchronized members, forwarding monothematic assets into closed family and community groups ($K \leq 256$; Fig. 4d) with high interpersonal trust ($\tau_{\text{WA}} = 0.85$, $2.8\times$ conversion). Mass In-Person Capital Rallies (T5, $+0.88\%$, $+1.05$M votes) assemble 1.80M citizens across open-air rallies in 27 state capitals combined with municipal free-transit decrees to produce physical turnout locking ($A_{i,L} > \tau_{\text{turnout}}$, Modality $v = 4$; Fig. 3a), while generating high-arousal aerial crowd imagery that shatters opponent claims of street supremacy. Finally, 24-Hour Live Stream Saturation (T6, $+0.80\%$, $+0.95$M votes) maintains an uninterrupted 24-hour broadcast event during the final 48 hours that captures 65% of public digital attention, creating a communicative ceiling that compresses rival blitz visibility below 20% and prevents late-stage vote inversion.

The dose-response and iso-vote equivalence landscape (Fig. 6d) illustrates tactical fungibility across the intervention portfolio. Along the benchmark iso-vote contour of $+1.36$M votes ($+1.14\%$ valid share), an identical outcome is achieved by an 88% asphyxiation compliance rate (88% intensity), a 24-hour agenda concentration of HHI $= 0.79$ (72% intensity), a loss-aversion multiplier of $\lambda_{\text{loss}} = 1.98\times$ (49% intensity), or 10.7M synchronized WhatsApp members (72% intensity) (Fig. 6d).

The cumulative addition of counter-tactics reveals a decisive **threshold principle** (Fig. 6e): starting from the Round 1 baseline (45.16%), Algorithmic Asphyxiation (46.30%), Strict Monothematism (47.65%), and Material Loss Framing (49.07%) steadily close the gap. However, the pivotal 50% majority threshold is crossed only upon adding Tactic 4 (Molecular WhatsApp Synchronization, reaching 50.12%), while Mass Capital Rallies (51.00%) and the 24-hour Live Stream (51.80%) lock the victory firmly outside the survey margin of error. Together, these empirical calibrations establish that democratic resilience against computational propaganda requires neither censorship nor passive fact-checking, but rather the structural operationalization of algorithmic dynamics in defense of popular majorities.

Discussion

The findings reported here provide a systemic resolution to a central paradox of contemporary democratic politics: how well-funded, institutionally entrenched political incumbents possessing verified popular majorities can be abruptly derailed by computational propaganda, and how such digital offensives can be systematically defeated. By grounding our analysis in a calibrated digital twin of the 2026 Brazilian presidential contest, we have isolated the precise structural and cognitive vulnerabilities that mediate late-campaign volatility in polarized polities.

Our results demonstrate that late-campaign volatility is fundamentally governed by a profound cognitive asymmetry between candidate admiration and candidate aversion. While positive voter admiration accumulates gradually and decays rapidly ($\lambda_A = 0.12, \delta_A = 0.06$), candidate rejection is acquired with extreme rapidity and displays strong hysteresis ($\lambda_R = 0.14, \delta_R = 0.04$). This $3.5\times$ kinetic asymmetry implies that negative emotional shocks induce persistent psychological scars that cannot be rapidly erased by conventional policy discourse. When combined with decision utilities that penalize candidate aversion more heavily than they reward positive affinity ($\beta_R = 1.45 > \beta_A = 1.15$), targeted moral panics and corruption allegations deployed during the final 48 hours of a campaign do not need to convert ideological partisans; they need only elevate out-group rejection among cross-pressured swing voters to suppress turnout or induce defensive vote defection.

Crucially, our multi-layer network architecture demonstrates why traditional institutional defenses systematically fail against these offensives. Legacy broadcast television and radio operate under a fundamental separation of timescales ($\tau_{\text{turnaround}} \gg \tau_{\text{social}}$). While institutional newsrooms and regulatory courts debate editorial standards and legal compliance over cycles of days or weeks, algorithmic social media platforms and encrypted messaging networks operate on hourly lifecycles. Furthermore, platform recommendation engines systematically reward controversial content along the diagonal of dual-valence moral outrage ($R_m = \text{Likes}_m + \omega \cdot \text{Dislikes}_m$). Under this algorithmic incentive structure, well-intentioned defensive counter-speech, outrage debunking, and quote-dunking paradoxically provide the very controversy signals required by recommendation engines to propel opponent disinformation into unaligned discovery feeds.

The decisive success of the second-round Janonismo Cultural counter-offensive establishes that democratic

resilience in the digital age requires an active structural posture rather than passive legalism. By enforcing algorithmic asphyxiation (T1), allied networks starved opponent content of cross-cluster friction, dampening rival reach by depriving platforms of engagement telemetry. By enforcing strict 24-hour monothematism (T2), allied networks exploited mathematical convexity in platform trending curves ($V(\mathbf{x}) \propto \sum x_k^2$), quadrupling their virality over fragmented messaging. By replacing abstract defenses of democratic institutions with visceral material-loss framing (T3), the campaign engaged Kahneman–Tversky loss aversion ($\lambda_{\text{loss}} = 2.25\times$), mobilizing lower-SES citizens whose economic vulnerability made them sensitive to welfare rollbacks. Finally, by synchronizing closed dark-social forwarding (T4) with physical capital rallies (T5) and an unbroken 24-hour livestream saturation ceiling (T6), the campaign bypassed algorithmic gatekeeping, locked in voter turnout, and definitively inverted the affective balance of the election.

Several limitations of our framework should be noted. Although our agent-based model is grounded in national census demography, detailed electoral tracking surveys, and empirical social media metrics, our simulations necessarily abstract interpersonal communication into discrete metric concept spaces and stylized network topologies. Future work should incorporate large language model (LLM) agents capable of generating and interpreting unconstrained textual and visual discourse. Moreover, while our analysis focused on the institutional and media architecture of Brazil, the core mechanisms identified here, namely dual-valence algorithmic curation, cognitive negativity bias, and dark-social cascade propagation, are universal features of contemporary information networks. Extending this multi-layer digital twin framework to other democratic contexts, such as the United States, India, and the European Union, will provide deeper cross-national insight into the design of platform architectures and democratic safeguards capable of preserving electoral integrity in an era of algorithmic warfare.

Methods

Concept space geometry and cognitive coupling

Citizens are embedded in a D-dimensional continuous concept hypercube $\Omega = [-1, 1]^D$ ($D = 4$), where dimensions represent economic/fiscal policy ($d = 0$), moral/social customs ($d = 1$), environmental regulation ($d = 2$), and public security ($d = 3$) (Supplementary Note 1). The normalized distance between any two vectors $\mathbf{u}, \mathbf{v} \in \Omega$ is defined as

$$d(\mathbf{u}, \mathbf{v}) = \frac{1}{\sqrt{D}} \|\mathbf{u} - \mathbf{v}\|_2 \in [0, 2], \tag{1}$$

such that the distance of any point $\mathbf{u}$ from the ideological origin $\mathbf{0}$ lies in the unit interval $E(\mathbf{u}) \equiv d(\mathbf{u}, \mathbf{0}) \in [0, 1]$. Ideological cross-issue interdependence is modeled via a row-normalized cognitive coupling matrix $\mathbf{C} = [C_{dd'}] \in \mathbb{R}^{D\times D}$ satisfying decaying off-diagonal constraints (Supplementary Note 1).

Demographic sampling and behavioral heterogeneity

Each citizen $i \in \{1, \ldots, N\}$ is assigned an age $a_i \in [16, 90]$ sampled from the empirical IBGE 2022 Census pyramid with Gaussian kernel smoothing ($\sigma_a = 1.35$ years), yielding normalized age $\tilde{a}_i = (a_i - 16)/74 \in [0, 1]$. Socioeconomic status is drawn from $\text{SES}_i \sim \text{Beta}(2.0, 3.0)$. Each agent maintains two coupled attitudinal states: an internalized slow-moving conviction vector $\mathbf{s}_i(t) \in \Omega$ and a fast-moving publicly expressed opinion vector $\mathbf{x}_i(t) \in \Omega$. Conviction extremity is defined as $E_i(t) = \|\mathbf{s}_i(t)\|_2/\sqrt{D} \in [0, 1]$. Heterogeneous psychological parameters governing social susceptibility α_i, bounded confidence threshold ϵ_i, attitude uncertainty $u_i(t)$, and confirmation bias b_i are conditioned on age, SES, and conviction extremity (full parameter distributions and functional specifications in Supplementary Note 1).

Multiplex interpersonal network construction

Interpersonal encounters unfold across a five-layer multiplex network $\mathcal{M} = \{\mathcal{L}_1, \ldots, \mathcal{L}_5\}$ stored in Compressed Sparse Row (CSR) format: Layer 1 (Household cliques, size distribution calibrated to 2.79 residents/household, weight $w_1 = 0.36$); Layer 2 (Workplace and school circular lattice peers, $\langle k^{[2]} \rangle = 10$, $w_2 = 0.26$); Layer 3 (Religious congregations and community groups, $\langle k^{[3]} \rangle = 8$, $w_3 = 0.14$); Layer

4 (Small-world friendship circles with Watts–Strogatz rewiring $p_{\text{ws}} = 0.15$, $\langle k^{[4]} \rangle = 6$, $w_4 = 0.14$); and Layer 5 (Erdős–Rényi transit weak ties, $\langle k^{[5]} \rangle = 4$, $w_5 = 0.10$). Composite edge weights $W_{ij} = \sum_{\ell=1}^{5} w_\ell A_{ij}^{[\ell]} / \sum_k \sum_\ell w_\ell A_{ik}^{[\ell]}$ yield a mean interpersonal degree of $\langle k_{\text{comp}} \rangle \approx 30.8$ ties per citizen (Supplementary Note 1).

Two-timescale opinion and conviction dynamics

At each discrete step $\Delta t = 0.15$, citizen i interacts with peers $j \in \mathcal{N}_i$ satisfying $d_{ij}(t) = \frac{1}{\sqrt{D}} \|\mathbf{x}_j(t) - \mathbf{x}_i(t)\|_2 \le \epsilon_i + 0.5\, u_i(t)$. Effective peer weights combine relative agreement uncertainty overlap $\phi_{ij}^{\text{RA}}(t)$ and biased assimilation cosine similarity $\phi_{ij}^{\text{BA}}(t)$:

$$\tilde{w}_{ij}(t) = W_{ij}\big(0.4 + 0.6\, \phi_{ij}^{\text{RA}}(t)\big)\phi_{ij}^{\text{BA}}(t)\, \mathbb{I}\big(d_{ij}(t) \le \epsilon_i + 0.5\, u_i(t)\big). \tag{2}$$

The interpersonal social shift $\Delta \mathbf{x}_i^{\text{soc}}(t) = \sum_j \tilde{w}_{ij}(\mathbf{x}_j - \mathbf{x}_i) / \sum_j \tilde{w}_{ij}$ updates current opinions through the Friedkin–Johnsen formulation:

$$\mathbf{x}_i^{\text{int}}(t) = \text{clip}_{[-1,1]^D}\Big(\mathbf{x}_i(t) + \Delta t\big[(1-\alpha_i)(\mathbf{s}_i(t) - \mathbf{x}_i(t)) + \alpha_i \mathbf{C} \Delta \mathbf{x}_i^{\text{soc}}(t)\big] + \boldsymbol{\xi}_i(t)\Big), \tag{3}$$

with noise $\boldsymbol{\xi}_i(t) \sim \mathcal{N}(\mathbf{0}, \sigma_\xi^2 \mathbf{I}_D)$ ($\sigma_\xi = 0.012$). When an exogenous fact impulse $\mathbf{b}_{\text{fact}}(t)$ is present, opinions pull toward it with sensitivity $\gamma_i(t) = \gamma_0(1.30 - 0.72E_i)(1.18 - 0.48\tilde{a}_i)(0.85 + 0.30\,\text{SES}_i)$, yielding intermediate opinion $\tilde{\mathbf{x}}_i(t)$. Opinions are elastically bounded around convictions within radius $\Delta_{\max,i}(t)$ via radial hyperbolic-tangent tethering:

$$g_i^{\text{eff}}(t) = \Delta_{\max,i}(t) \tanh\left(\frac{\frac{1}{\sqrt{D}}\|\tilde{\mathbf{x}}_i(t) - \mathbf{s}_i(t)\|_2}{\Delta_{\max,i}(t)}\right), \tag{4}$$

$$\mathbf{x}_i(t+1) = \text{clip}_{[-1,1]^D}\left(\mathbf{s}_i(t) + \frac{g_i^{\text{eff}}(t)}{g_i(t) + 10^{-12}}\big(\tilde{\mathbf{x}}_i(t) - \mathbf{s}_i(t)\big)\right). \tag{5}$$

Convictions adapt slowly via crystallization rate $\eta_{\text{cryst},i}(t) = \eta_0[1 + (t/\tau_c)^{1.15}]^{-1}(1.25 - 0.75\tilde{a}_i)(1.22 - 0.68E_i)(1.10 - 0.25\,\text{SES}_i)$ and sustained-tension rate $\eta_{\text{sust},i}(t) = 0.28\eta_0(g_i^{\text{eff}}/\Delta_{\max,i})^2(1.18 - 0.58\tilde{a}_i)(1.20 - 0.62E_i)$ ($\eta_0 = 0.0024$, $\tau_c = 280$ steps):

$$\mathbf{s}_i(t+1) = \text{clip}_{[-1,1]^D}\Big(\mathbf{s}_i(t) + \big[\eta_{\text{cryst},i}(t) + \eta_{\text{sust},i}(t)\big]\big(\mathbf{x}_i(t+1) - \mathbf{s}_i(t)\big)\Big). \tag{6}$$

Traditional mass media and editorial tendentiousness

Media conglomerates $g \in \{1, \ldots, M\}$ operate across five modalities $v \in \{0, 1, 2, 3, 4\}$ (TV, Radio, Press, Outdoor, Rallies) with empirical demographic reach probabilities $P_v(a_i, \text{SES}_i)$ (Supplementary Note 2). Outlets project factual events $\mathbf{f}_k(t)$ toward their editorial ideological centers $\mathbf{m}_g \in \Omega$ through tendentiousness parameter $\tau_g \in [0, 1]$:

$$\mathbf{y}_{k,g,v}(t) = (1 - \eta_{g,v})\, \mathbf{f}_k(t) + \eta_{g,v}\, \mathbf{m}_g + \boldsymbol{\xi}_{k,g,v}, \qquad \eta_{g,v} = \text{clip}_{[0,1]}(\tau_g \mu_v), \tag{7}$$

with baseline reach μ_v and noise $\sigma_{\text{media}} = 0.03$. Active notices decay exponentially as $I_v(t+1) = I_v(t)\exp(-\delta_v)$. Citizens select active vehicles via distance-weighted softmax exposure and update opinions within credibility-expanded confidence bounds $d(\mathbf{x}_i, \mathbf{y}_v) \le \epsilon_i(1 + 1.10\,\gamma_v)$ (Supplementary Note 2).

Algorithmic platforms, engagement, and cascades

Digital media encompasses four platform modalities $p \in \{0, 1, 2, 3\}$ (Instagram/TikTok, YouTube/X, Facebook, WhatsApp). Content creators accumulate followers via preferential attachment $\Pi_k \propto k_i(0.65 + 0.70\tau_{\text{inf},k})$, generating power-law in-degrees $P(k) \sim k^{-1.15}$ (Gini 0.73). Feed algorithms recommend posts with probability $q_{ij} \propto (d(\mathbf{x}_i, \mathbf{y}_j) + \delta)^{-\beta}(1 + 1.5e_j)\ln(1 + \text{pop}_j)(0.5 + 0.5\ln(1 + k_{\text{inf}}))$ with homophily $\beta = 2.0$. Reactions branch into likes ($d \le \epsilon_{\text{like},i}$, assimilating toward $\mathbf{y}_j$) and dislikes/outrage ($d \ge \tau_{\text{dislike},i}$,

triggering backfire repulsion away from $\mathbf{y}_j$; Supplementary Note 3). Algorithmic relevance scores compound dual-valence engagement symmetrically:

$$R_m = \text{Likes}_m + \omega \cdot \text{Dislikes}_m, \qquad \omega = 1.0, \tag{8}$$

maximizing viral discovery along the diagonal of controversy (Supplementary Note 3).

Electoral decision calculus

Candidates $k \in \{L, F, O\}$ are positioned at fixed coordinates $\mathbf{c}_k \in \Omega$. Candidate affinity $A_{ik}(t)$ and aversion $R_{ik}(t)$ evolve with asymmetric parameters ($\lambda_A = 0.12, \delta_A = 0.06; \lambda_R = 0.14, \delta_R = 0.04$):

$$A_{ik}(t+1) = (1-\delta_A)\, A_{ik}(t) + \lambda_A \max\left(0, \frac{0.62 - d_{ik}(t)}{0.62}\right)^2, \tag{9}$$

$$R_{ik}(t+1) = (1-\delta_R)\, R_{ik}(t) + \lambda_R \min\left(1, \max\left(0, \frac{d_{ik}(t) - 0.62}{0.73}\right)^{1.35}\right) + \lambda_{\text{rej}}\, \mathbb{I}_{\text{exposed}}(i). \tag{10}$$

Citizen i evaluates overall utility:

$$U_{ik}(t) = 1.15\, A_{ik}(t) - 1.45\, R_{ik}(t) - 0.45\, d_{ik}(t) + \xi_{ik}(t), \qquad \xi_{ik} \sim \mathcal{N}(0, 0.03^2). \tag{11}$$

Ballot decisions follow a three-way calculus: citizens abstain ($C_i = -1$) when $\max_k A_{ik} < 0.14$ or $\min_k R_{ik} > 0.60$; cast blank/null ballots ($C_i = -2$) when $|U_{i0} - U_{i1}| < 0.10$ and $\max_k R_{ik} < 0.52$; or cast valid votes for $C_i = \arg\max_k U_{ik}$ (Supplementary Note 5).

Simulation protocol and statistical evaluation

Simulations were executed across $S = 100$ independent Monte Carlo realizations at population scale $N = 100{,}000$ using JIT-compiled Numba kernels (Supplementary Note 5). Metrics include multidimensional Esteban–Ray polarization P_{ER} ($\alpha_{\text{ER}} = 1.3$, $\kappa_{\text{ER}} = 1.0$), Sarle's bimodality coefficient BC, cross-issue Pearson correlation alignment $|\bar{\rho}|$, and Roy's eigenvalue participation ratio $\text{PR}(\Sigma)$. All time-series trajectories report ensemble means with 95% confidence intervals ($\bar{\mu} \pm 1.96\, \hat{\sigma}/\sqrt{S}$) or full simulation ranges $[\min, \max]$.

Data availability

All empirical data supporting the findings of this study, including 2022 IBGE demographic census records, 2026 TSE official ballot counts, Datafolha longitudinal tracking survey archives, and social media volume telemetry from Bites, are publicly accessible and deposited in the study repository at https://github.com/cesarrennocosta/janones.

Code availability

The complete computational simulation engine, Numba-compiled array kernels, and experiment execution pipelines are available under an open-source MIT license in the study repository at https://github.com/cesarrennocosta/janones.

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